MIL-R-2470E
28 March 1980
SUPERSEDING
MIL-R-2470D
24 April 1975

MILITARY SPECIFICATION

REFRIGERATORS, MECHANICAL, MORTUARY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of self-contained mortuary refrigerator equipped with telescoping racks.

1.2 Classification. Mortuary refrigerators covered by this specification shall be of the following sizes and grades as specified (see 6.2):

Size 2 - 2 Cadavers
Size 5 - 5 Cadavers

Grade A - Stainless steel inner liner and outer shell.
Grade B - Aluminum inner and outer shell.
Grade C - Baked enamel on steel inner liner and outer shell.
Grade D - Stainless steel for exterior front, zinc coated steel on balance of exterior; vitreous porcelain enamel on steel for interior walls and ceiling, gray enamel on zinc coated steel for floor.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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SPECIFICATIONS

FEDERAL

BB-F-1421	-	Flurocarbon Refrigerants
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-A-596	-	Aluminum Alloy Permanent and Semipermanent Mold Castings
QQ-A-601	-	Aluminum-Alloy Sand Castings
QQ-B-654	-	Brazing Alloy, Silver
QQ-S-775	-	Steel Sheets, Carbon, Zinc-Coated
TT-E-489	-	Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces)

MILITARY

MIL-S-6872	-	Soldering Process, General Specification for
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing Of
MIL-P-12742	-	Primer Coating, Phenolic, Water Immersible

STANDARDS

FEDERAL

FED-STD-406	-	Plastics, Methods of Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

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UNDERWRITERS LABORATORIES, INCORPORATED (UL)

Standard No. 471 - Commercial Refrigeration and Freezers 15 Sept 75
Electrical Materials Construction List

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062; 207 East Ohio Street, Chicago, IL 60611; Walt Whitman Road, Melville, L.I. NY 1655 Scott Blvd., Santa Clara, CA 95050; or 2602 Tempa East Blvd., Tampa, FL 33619.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies of Handbook H28 should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium - Nickel Sheet and Strip
B 86 - Zinc, Alloy Die Castings

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNITED STATES DEPARTMENT OF COMMERCE

Standard PS-1 - Softwood Plywood Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Refrigerators delivered under this specification shall be the manufacturer's standard commercial catalog product, except for any changes necessary to comply with specification requirements.

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3.2 Safety standard. All mortuary refrigerators shall conform to the applicable requirements of UL Standard No. 471.

3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the mortuary refrigerators he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 471. Acceptable evidence of meeting the requirements of this standard shall be the UL label, a listing in the UL "Electrical Materials Construction List" or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the mortuary refrigerators offered have been tested and conform to the applicable requirements of UL Standard No. 471.

3.3 Materials. Materials not definitely specified shall be of the quality normally used for the purpose by the contractor. Unless otherwise specified herein, commercial tolerances shall apply to material thicknesses and diameters (see 6.6).

3.3.1 Lumber. All lumber shall be select grade hardwood.

3.3.2 Plywood. All plywood shall be group I, exterior type, grade B-C sanded on two sides and shall conform to PS-1.

3.3.3 Preservative, wood. Wood preservative shall be applied to all wood members in accordance with the manufacturer's commercial practice.

3.3.4 Insulation.

3.3.4.1 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane foam having a manufacturer's commercial rated "K" factor of not more than 0.15 (0.26 W/m.k). The insulation shall have a density of not less than 1.6 pounds per cubic foot (26 kg/m^3), when tested as specified in 4.4.8.

3.3.4.2 Semirigid. Semirigid insulation shall be B fiber or finer industrial type fiberglass having a manufacturer's commercial rated "K" factor of not more than 0.25 (0.43 W/m.k) and a manufacturer's commercial rated density of at least 1.0 but not more than 3.0 pounds per cubic foot (16-48 kg/m^3) and shall be held together with a thermosetting binder.

3.3.4.3 Rigid. Rigid insulation shall be preformed expanded polystyrene made from either beads or from the resin by the extrusion process with a manufacturer's commercial rated "K" factor of not more than 0.25 (0.43 W/m.k) and a minimum density of 1.8 pounds per cubic foot (29 kg/m^3) (see 4.4.8).

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3.3.5 Plastics. Plastics shall be virgin material except for reclaimed sprues, runners, or rejects from the same molding operation. The term virgin shall not preclude the use, in paint and plastic compounds, of reground ingredients from the same operation when this use is in accordance with the best commercial practice and when the materials reground have never before been fabricated. The term "fabricated" in the case of plastic compounds, is defined as the manufacture of plastic products by the assembly of parts or standard engineering shapes by cutting, drilling, tapping, or similar machine operations, and fastening the parts or shapes together with mechanical devices or adhesives. The exclusion of such "fabricated" materials is intended to prevent the admixture of foreign matter during regrinding.

3.3.6 Thermoplastic materials.

3.3.6.1 Styrene. Styrene type plastics shall be made by the polymerization of styrene or shall be styrene-rubber plastics. Styrene-rubber plastics shall consist of at least 50 percent styrene plastics combined with rubbers and other compounding ingredients. Styrene type plastics shall have an Izod impact resistance of at least 0.70 foot-pounds (ft-lbs) (0.95 J) and when used for ornamentation, the Izod impact resistance shall be at least 0.25 ft-lbs (0.34 J) (see 4.4.10.1).

3.3.6.2 Thermosetting plastic. Thermosetting plastic shall be phenolic or melamine plastic. Molded parts shall have a flexural strength of not less than 7500 pounds per square inch (psi) (52 MPa) and laminated parts shall have a flexural strength of not less than 10,000 psi (69 MPa). Molded parts shall have a maximum water absorption of 0.8 percent and laminated parts shall have a maximum water absorption of 4.0 percent (see 4.4.10.2).

* 3.3.7 Bolts, nuts, screws, and washers. Bolts, nuts, screws, and washers used for mounting components shall be of the quality and in the sizes, types and materials recommended by the contractor. Threads shall conform to the National Bureau of Standards Handbook H28. Steel bolts, nuts, screws, and washers shall be plated or coated to resist corrosion. Brass bolts, nuts, screws, and machine screws shall not be used in fastening aluminum parts. All exposed screws and bolts shall be of stainless steel.

3.3.8 Refrigerant. The refrigerant shall conform to type 12, 22, or 502 of BB-F-1421.

3.3.9 Primer. Primer shall be phenolic type conforming to MIL-P-12742.

3.3.10 Enamel. The enamel shall be glossy white conforming to class B of TT-E-489.

3.3.11 Steel.

3.3.11.1 Sheet steel for porcelain enameling. Carbon sheet steel for porcelain enameling shall be (1) a commercial grade vitreous-enameling type ingot sheet,

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(2) metal sheets of sheet iron stabilized with the addition of titanium when one coat titanium process porcelainizing is used, or (3) may be any of the other metals suitable for two-coat porcelain enameling provided the sheets are plated with a nickel "flash" prior to being porcelain enameled.

3.3.11.2 Zinc-coated sheet steel. Zinc-coated carbon steel sheets shall conform to type I, class d of QQ-S-775.

3.3.11.3 Stainless steel plates, sheets, and strips. Stainless steel plate, sheet, and strip shall conform to ASTM Standard A 176.

3.3.12 Aluminum.

3.3.12.1 Aluminum plate and sheet. Aluminum plate or sheet shall conform to temper H14 of QQ-A-250/2.

3.3.12.2 Castings. Aluminum castings shall be sand or permanent-mold fabricated. When sand castings are supplied, they shall conform to alloy 108, 122, 142, 195, or 214 of QQ-A-601. When permanent mold castings are supplied, they shall conform to alloy number B195 of QQ-A-596.

3.3.13 High-pressure die cast zinc. When high pressure, die cast zinc is used, the zinc base alloy die-casting shall be manufactured and finished in conformance to ASTM Standard B 86. The alloy employed shall be either AG40A or AC41A. The castings shall be made smooth by fileing and tumbling before plating is applied. Each zinc-base die casting shall be plainly marked with the manufacturer's name, brand, or trademark in a location and manner that will not affect the use or appearance of the item.

* 3.3.14 Silver-base-brazing alloy. Silver base-brazing alloy shall conform to QQ-B-654.

3.4 Design. The refrigerators shall be capable of holding two cadavers in a size 2 and five cadavers in a size 5 (see 1.2). The refrigerators shall be of any of the four grades, A, B, C, and D, as specified. The refrigerator shall be free-standing when assembled and shall consist essentially of the cabinet, carriage assembly, and refrigeration system. The refrigerator shall be of sectional design and of a size to permit passage of sections through a 32 - by 80-inch (0.8 by 2 m) doorway. Any design incorporating the assembly of panels shall have mating surfaces of tongue and groove joint construction or butt joint construction designed to seal tightly and give complete rigidity to the assembled refrigerator (see 4.4.1). The refrigeration system shall automatically maintain temperatures in the refrigerator ranging from 32°F to 34°F (0°C - 1°C) with the refrigerating equipment running 80 percent of the time (see 4.4.3). The structural framing shall be designed to provide support for the fully loaded refrigerator. Each mortuary refrigerator, when assembled for operation, shall be fully self-supporting. The refrigerating equipment shall be enclosed in the top section of

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the refrigerator with adequate ventilating grills or louvers. The condensing unit compartment shall be separated from the refrigerated area by a space equal to the minimum thickness of insulation permitted for sections of the refrigerator. A vapor-sealing material shall be used between breaker strip and inner liner and outer shell. All component parts of a refrigerator unit of a given size and grade shall be interchangeable with parts of other units of the same size and grade furnished under this specification on any one contract or order.

* 3.4.1 Dimensions. The mortuary refrigerators shall be fabricated to the dimensions for the sizes given in table I.

TABLE I. Dimensions-inches, (mm)

Size	Width overall maximum	Depth overall maximum	Height overall maximum	Door opening inside (minimum)
2	40 (1016)	96 (2438)	80 (2032)	27 (686) wide by 22 (559) high
5	73 (1854)	96 (2438)	88 (2235)	27 (686) wide by 22 (559) high

3.5 Construction. The joints of all sections of the mortuary refrigerator shall be provided with gaskets to prevent the infiltration of air and moisture, and all sections shall be sealed to prevent infiltration of moisture into the insulated spaces. The space between the inner liner and outer shell shall be filled with insulation. All wood members used in the fabrication of the refrigerators shall be treated with material specified in 3.3.3. Necessary trimming or resurfacing of completely milled, treated, wood members will be permitted to secure the required fit provided surfaces are thoroughly coated prior to assembly with wood-preservative specified herein. All sections and doors shall be covered with one-piece metal skins inside and outside. Continuous metal paths between inner liner and outer shell shall be held to minimum.

3.5.1 Soldering. The soldering process shall be as described in MIL-S-6872.

3.5.2 Handholes. Handholes shall be provided and all handholes shall open to the cabinet interior to permit assembly of refrigerator sections in confined or limited areas. The handholes shall be sealed to prevent any insulation in the cabinet from sifting into the handholes or any moisture from infiltrating the cabinet insulation. Upon completion of assembly, all handholes, shall be packed with waterproof flexible plugs designed to completely fill the handholes. The plugs shall be packed with the insulation specified in 3.3.4. All handholes shall be provided with a cover fabricated of the same material used for inner liners for the inside and outer shells for the outside (see 3.5.6.1). The cover shall be vapor-sealed and held securely in place by screws or spring clips.

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3.5.3 Fasteners. The fastening devices used to retain the assembled sections of refrigerators shall be either a lag bolt panel type, a threaded metal pod type, or a cam action type. Threads shall conform to the National Bureau of Standards Handbook H28. The pod shall be securely attached to framing members so as not to require use of tools to hold the pod in place during refrigerator assembly. Lag bolts shall not bottom in pod when assembled in order to provide for further tightening if required, and shall be protected from cutting into wood framing members by suitable washers. The cam action type panel fasteners in the retracted position shall not protrude from the panels. When cam action type fasteners are supplied, two sets of erection tools compatible with the fasteners shall be furnished with each refrigerator.

3.5.4 Rack assembly. Each cadaver compartment shall be equipped with a mortuary rack consisting of rack frame, carriage assembly, and removable tray. The rack assembly shall be constructed so that the tray shall clear the door sills as closely as possible without contact. Stationary frames shall be provided as supports for the carriage assembly. The stationary frames shall be fabricated of the zinc-coated steel sheet specified in 3.3.11.2 and in a minimum thickness of 0.1233 inch (3.12 mm). The frames shall bear on reinforced or on supported cabinet sections to prevent damage to inner liner or outer shell. The rack assembly shall show no permanent distortion when loaded with 500 pounds (227 kg) and tested as specified in 4.4.2.

3.5.5 Carriage assembly. The carriage shall be of two-piece telescopic construction made of specially formed steel shapes and channels. The overall length of carriage assembly shall be not greater than 86 inches (2184 mm) and shall be able to pass through door openings having at least 27 inches (686 mm) of clearance width without striking the door jambs. The inside end of the upper carriage shall be equipped on each side with rollers or wheels designed to travel in the channels of the lower carriage, or bed, and support the rear end of the carriage. The extended tray and carriage shall maintain a 500 pound (227 kg) load without other support and without sagging or racking (see 4.4.2). The outer end of carriage shall be equipped with an automatic trip latch and handle for locking carriage and tray in position inside the refrigerator, and designed to allow tray only, or tray and carriage to be withdrawn. The fabricated carriage shall be of zinc-coated steel sheet specified in 3.3.11.2.

3.5.5.1 Lower carriage. The lower carriage or bed shall be constructed of longitudinal channel-type side members with cross members located at each end. The channels shall be fabricated of the zinc-coated steel sheet specified in 3.3.11.2 which shall not be less than 0.108 inches (2.7 mm) in thickness. The lower carriage shall ride on a minimum of six rollers or flanged wheels as specified in 3.5.5.3. Bumpers shall be provided at the rear end of the carriage to absorb the shock of closing.

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3.5.5.2 Upper carriage. The upper carriage, or frame, shall be designed and constructed to support the removable tray. The carriage shall be fabricated of the zinc-coated steel sheet specified in 3.3.11.2 which shall not be less than 0.1084 inches (2.7 mm) in thickness. The upper carriage shall be fitted with a minimum of four, full width, tray rollers with roller and stud or wheels as specified in 3.5.5.3. The rollers or wheels shall support the tray and permit end movement sufficient for withdrawal of the entire tray outside of the refrigerator. Bumpers shall be provided at the rear end to absorb any shock of telescoping the upper carriage assembly.

* 3.5.5.3 Rollers or wheels. The nylon rollers or corrosion resisting wheels of the carriage shall have ball bearings or bronze bushings for bearing surfaces mounted on studs or axles fabricated of stainless steel specified in 3.3.11.3.

3.5.5.4 Trays. Trays shall be fabricated of one-piece drawn stainless steel specified in 3.3.11.3 with nominal thickness of 0.0312 inch (.79 mm) and in an overall minimum dimension of 22 inches (559 mm) wide by 83 inches (2108 mm) long by 2-3/4 inches (70 mm) deep. The trays shall be removable and designed to support a uniformly distributed load of 500 pounds (227 kg) (see 4.4.2) and constructed with the sides and ends flared. The frame for the trays shall be constructed of continuous one-piece stainless steel seamless tube 1-inch (25.4 mm) outside diameter. The sides of the tray shall be formed neatly around the frame. The trays shall be smooth surfaced without cracks or crevices. The trays shall be provided with handles or hand holes at each end for lifting the tray. Handles, when used, shall be part of the frame and extend across the full width of the tray at approximately 3 inches (76 mm) beyond the ends of the sheet steel flare.

3.5.6 Refrigerated cabinet. The refrigerated cabinet shall consist of a frame with inner liner and outer shells between which the insulation shall be placed. The insulated cabinet shall be designed and constructed to prevent any displacement of the inner liner with respect to the outer shell, and to provide firm support for the rack assemblies. The insulation shall be the types specified in 3.3.4, and shall be installed so that it will not pack or settle in shipping, testing, or operation. The cabinet shall be designed so that insulated compartments shall show no signs of condensation on the outside of the compartments or doors (see 4.4.4). Electric heating in doors, door mullions, and sills may be used to prevent condensation.

3.5.6.1 Inner liners and outer shell. The inner liners and outer shells of grade A refrigerators shall be fabricated of the stainless steel specified in 3.3.11.3 and shall have a minimum thickness of 0.031 inch (0.79 mm). Grade B refrigerators shall be fabricated of the aluminum specified in 3.3.12.1 and shall have a minimum thickness of 0.040 inch (1 mm). Grade C refrigerators shall be fabricated of the zinc-coated steel sheet specified in 3.3.11.2 and shall have a minimum thickness of 0.0396 inch (1 mm). Grade D refrigerators shall be fabricated of the materials indicated in 1.2. The nominal thickness of stainless steel and zinc-coated steel shall be the same as specified for grades A and C refrigerators. The exterior bottom of cabinet shall be phosphatized galvanized steel. The minimum thickness for porcelain enameling steel shall be 0.0359 inch (0.9 mm).

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3.5.6.2 Breaker strips. Breaker strips shall be used to securely join the inner liner with the outer shell around all door openings and shall provide a smoothly finished appearance. Breaker strips shall be fabricated of plastic specified in 3.3.6 and shall be attached to the inner liner and outer shell with screws as specified in 3.3.7. Screws shall be of a type easily removed with a screwdriver. Where breaker strip ends meet door-opening corners, cornice clips of stainless steel specified in 3.3.11.3 shall be installed without corner clips if joints so connected are smooth, water-proof, and easily cleaned. Breaker strips may be black, white, or colored according to the manufacturer's current design.

3.5.6.3 Doors. The doors shall be metal covered and the space between framing members and skins shall be of insulation specified in 3.3.4. The inner liners and outer shells shall be constructed of the same material as the class of refrigerator specified. The door shall overlap the cadaver compartment opening edges and overlapping section shall have attached thereto a gasket, as specified in 3.5.6.3.1, to seal the opening against air and moisture infiltration when door is closed. Each door shall be supported by two hinges and secured, when in closed position, by a latch-and strike-fastener with sufficient force to affect a gasket seal (see 4.4.7). The door shall have a card holder mounted to outside surface, fabricated of stainless steel or finished to match cabinet hardware. Card holder shall accommodate a name card 2-1/4 inches by 2-7/8 inches (57 by 73 mm).

3.5.6.3.1 Gaskets. The doors shall be furnished with replaceable cushion type one piece gaskets of synthetic rubber or plastic. Plastic gaskets shall comply with the requirements for plastics in 3.3.5. The gaskets shall be close fitting and symmetrically installed to provide air-tight sealing.

3.5.6.4 Drain. A water-sealed, bell-type, trap drain with removable strainer shall be located beneath the interior bottom of cabinet at the center rear, rigidly fastened thereto, and permanently sealed at inner liner and outer shell to prevent entry of water or water vapor into the insulation. The trap shall be provided with a removable, perforated disc-type cover. The strainer shall be of the perforated disc-cover type and shall provide an outlet opening for a minimum 3/4 inch Iron Pipe Size (IPS) drain pipe. The material used for the trap, cover, pipe, tee, plug and strainer shall be brass or cast aluminum. The drip-pan and drain pipe shall be of stainless steel, aluminum, brass, polyethylene tubing, or copper tubing and shall be either of large enough diameter for easy cleaning or shall be readily removable for cleaning.

3.5.6.5 Lighting. Each refrigerator shall be equipped with an interior light or lights mounted in a vaporproof fixture on the interior of the front section of the refrigerator. Lights shall be controlled by a toggle switch on the refrigerator exterior front. The toggle switch shall be equipped with a pilot light to indicate when the light is turned on. All wiring shall be within the same refrigerator section. The entire installation within the cabinet and panel shall be vapor proof. Provision shall be made for connecting the lighting circuit to an outside source of electrical current.

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3.5.6.6 Hardware. The latch and strike assembly shall be of the adjustable type, constructed of cast or forged brass, high pressure die-cast zinc, or malleable iron and shall be polished chrome or nickel finish. When die-cast latch and strike are furnished, the latch shall be forged brass and the sticker and tongue shall be extruded brass or forged steel. A latch and striker assembly shall be provided on each compartment door with separate locking facilities capable of taking a 1-3/4 inch (44.5 mm) padlock. The hinges shall match the latch in finish and material. The hinges shall be the ball-bearing type. Doors opening toward adjacent doors shall be equipped with 135 degree stop hinges.

3.5.6.7 Thermometer. A remote indicating dial thermometer shall be mounted on the outside front surface with the temperature-sensing element located near the center of the refrigerated space but not in the direct path of airflow. The temperature-sensing element shall be affected by air temperature only. The thermometer shall have a white face and dark figures with a 2-inch, plus or minus 1/2-inch (38 to 64 mm) diameter dial. The dial shall be incased in a chromium plated or brass case. The thermometer shall be graduated in the Fahrenheit system and shall indicate temperatures from at least 20°F to 60°F. The thermometer shall be adjustable to compensate for any misalignment resulting from temperature extremes of storage and transportation.

* 3.6 Refrigeration system. Unless otherwise specified (see 6.2), the refrigeration system shall be designed to operate on a nominal 120-volt, 60-hertz (Hz), 1-phase electric power system. The system shall be self-contained, hermetic type, either thermally or mechanically sealed, or open type as specified (see 6.2), and designed for use with the refrigerant specified in 3.3.8. The refrigeration system shall be furnished complete with 1/2 horsepower (0.3 kW) compressor for size 2 and 3/4 horsepower (0.6 kW) for size 5, air-cooled condenser, unit cooler assembly, receiver, valves, gages, drier, motor, moisture and liquid indicator, and all necessary refrigeration and safety controls applicable to the refrigeration system. Provision shall be made for uniform air distribution within the refrigerated space. All components and piping containing refrigerant shall be gas tight (see 4.4.6). A thermal overload protective device with manual reset shall be supplied with the motor. The thermal overload protective device shall be rated to open the electrical circuit before the windings exceed the temperature rise for which the motor is rated. Fuses are not acceptable in lieu of the thermal overload protective device. Each assembled refrigeration system shall be dried, evacuated, and charged with operating requirements of refrigerant specified in 3.3.8 and oil recommended by compressor manufacturer. The moisture content of the refrigerant shall show dry on the moisture-liquid indicator (see 4.4.5). Framed ventilating grilles or louvers shall be installed to cover all access openings in the condensing unit inclosure, and shall be fabricated of the same material and finish as the shell for the class of refrigerator specified, and shall be attached to the cabinet in a manner to facilitate cleaning and replacement of equipment. The grilles and louvers shall be designed so that the openings consist of 75 percent of the area covered by the grille or louver.

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3.6.1 Unit cooler assembly. The unit cooler assembly shall be of the forced-convection type, located in the top section of the refrigerator and mounted to be easily accessible. Each unit cooler assembly shall be furnished complete with evaporator, fan and fan motor, heat interchanger, and all accessories, instruments, and controls necessary to provide and automatically maintain refrigerator temperature of 32°F to 34°F (1°C to 2°C) when operating at a mean temperature difference not exceeding 12°F (± 6.5°C). Unless otherwise specified (see 6.2), the fans shall be operated continuously and shall be designed for use on a nominal 120-volt, 1-phase, 60 Hz electrical power system. All components of the unit cooler assembly shall be contained within a protecting shroud of housing and provided with an integral drip pan containing a connection for a drain line. The evaporator coil joints shall be brazed with the material specified in 3.3.14.

3.6.2 Controls. Controls shall be provided to automatically maintain the specified cabinet temperatures. Motor cutout switches shall open the motor circuit when the discharge pressure becomes excessively high or when the suction pressure becomes excessively low. The automatic control equipment shall be positive in action, shall operate with a minimum of attention, and shall be adjustable through the temperature or pressure range required for the temperatures specified in 3.4.

3.7 Finish. Porcelain enamel or baked enamel finish shall be white, except that enamel for floor surfaces shall be gray. All sharp edges, corners, burrs, and slivers shall be removed.

3.7.1 Aluminum. When aluminum is used for the outer shells and inner liners it shall have either a stucco embossed finish or a bright, smooth finish free of stains or discolorations.

3.7.2 Stainless steel. Stainless steel fronts and sides of the outer shells and stainless steel door interiors and exteriors shall have a No. 4 finish. Stainless steel backs and tops of outer shells may have a No. 2B finish, stainless steel inner liners shall have a No. 4 finish.

3.7.3 Organic. Organic finishes shall be baked synthetic specified in 3.3.10 or acrylic enamel over a coat of primer specified in 3.3.9.

3.8 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with II B of MIL-STD-461 (see 4.4.9).

3.9 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacements. Access panel covers and plates shall be provided as required for component adjustment, repair, or replacement and shall be fastened with easily operated quick-disconnect fasteners or with screws.

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3.10 Marking for identification. Refrigerators shall be marked with the following information on a nonferrous metal plate at least 0.04 inch (1 mm) thick in accordance with MIL-STD-130. Marking shall be applied to the plate by metal stamp, vibro peening, acid, electric or electro-chemical etching, or embossing. The plate shall be fastened to the refrigerator with rivets, bolts, or screws.

(a) Nomenclature
(b) Size and grade (see 1.2)
(c) Federal Stock Number
(d) Manufacturer's model, type, or catalog number
(e) Serial number
(f) Contract or order number
(g) "US"
(h) Other information required by UL Standard No. 471.

3.11 Wiring diagram. A wiring diagram shall be glued or otherwise affixed in an accessible location on the refrigerator so that it may be read from the front of the refrigerator. The diagram shall be water-proof and shall portray each electrical component and its terminals by a sketch, outline, or name, and not by electrical symbol alone, showing the wiring of the refrigerator with the wires coded for identification.

3.12 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. Sheet metal or structural sections used in fabrication of the refrigerators shall not have kinks, bends, or ripples. Shearing and chipping shall be done accurately and cleanly on all portions of the work and shall be neatly finished. Any bolt and rivet holes shall be accurately and cleanly punched or drilled. Rivets, when used, shall be driven to completely fill the holes, with full sized heads neatly finished and in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they shall not work loose under vibration. Optional equipment, accessories, and piping shall be securely fastened and braced to prevent damage or loosening during transit or operation.

4. QUALTIY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Component testing.

4.2.1.1.1 Plastics test. The plastics test shall be performed as specified in 4.4.10.1 and 4.4.10.2. The inspection lot shall be the amount of each type of plastic received at one time for fabrication purposes. Three representative samples from each lot shall be tested. Failure of any test shall be cause for rejection of the inspection lot.

4.2.2 In-process inspection.

4.2.2.1 In-process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Wherever nonconformance is noted, correction shall be made to the items affected and to the operation.

(a) Vapor sealing as required by 3.4 and 3.5.2.
(b) Fasteners and lag bolts are installed to comply with 3.5.3.
(c) Insulation is installed as specified and wood members treated as specified.
(d) Organic finishes are primed as specified in 3.7.3.
(e) Workmanship is as specified in 3.12.

4.2.2.2 In-process testing.

4.2.2.2.1 Density of insulation test. The density test for insulation shall be performed as specified in 4.4.8. The inspection lot shall be the amount of material used in one day's production. The sample unit for this test shall be a 6-inch (152 mm) cube of insulation. Failure of the test shall be cause for rejection of the inspection lot.

4.2.3 End item inspection. The inspection lot shall consist of all refrigerators of the same size and grade offered for inspection at one time. The sample unit for the inspection shall be one refrigerator.

4.2.3.1 End item examination.

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4.2.3.1.1 Visual examination. Examination shall be made of the refrigerator for defects set forth in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examination	Defect	Classification Major	Minor
Finish	Not color specified		X
	Area of no film primer showing		X
	Not No. 4 where specified		X
	Not No. 2B where specified		X
	Sharp edges, corners, burrs, and slivers not removed	X	
Design, construction, and workmanship general (applicable to all components and assemblies)	Part missing	X	
	Any component fractured, split, punctured, dented, deteriorated, bowed, or malformed	X	
Handholes (in cabinet)	Missing, not sealed	X	
	Waterproof flexible plug too small or too large, handholes not insulated as specified		X
	Cover not same material, not vapor sealed		X
Fasteners	Not type specified	X	
Rack assembly	Does not clear door sills	X	
Carriage assembly	Not two-piece construction	X	
	Rollers or wheels out of channel		X
Trays	Not one piece construction	X	
	Sides and ends not flared		X
	Cracks or crevices visible		X
Breaker strips	Cornice clips not as specified		X

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TABLE II. Classification of defects (cont'd)

Examination	Defect	Classification Major	Minor
Design, construction, and workmanship general (applicable to all components and assemblies) (cont'd)			
Doors	Not types specified	X	
	Card holder not mounted where specified or as specified		X
Drain	Not type specified	X	
	Not located where specified		X
	Cover not type specified		X
	Drainpipe not readily removable for cleaning		X
Lighting	Not type specified, not mounted as specified	X	
	Switch not located where specified	X	
	No provision for connection to outside electrical source	X	
Thermometer	Not type specified	X	
	Not located where specified		X
	Not adjustable		X
Unit cooler assembly	Not type specified	X	
Marking for identification	Missing, incomplete, not legible, not specified type, not affixed or marked in prescribed manner, or not of the specified material		X

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4.2.3.1.2 Dimensional examination. Examination shall be made of the refrigerator for defects in dimensions. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.2.3.2 End item testing. One representative production refrigerator, including the refrigeration unit, shall be completely assembled and tested as specified in 4.4.1 through 4.4.7 and, when applicable, 4.4.9. Failure to pass any test shall be cause for rejection of the inspection lot. The refrigeration units of all other refrigerators of the inspection lot shall be tested as specified in 4.4.3, 4.4.4, and 4.4.5. The refrigeration units may be tested on the assembled representative production refrigerator. Failure of the refrigeration unit to pass any of the three tests shall be cause for rejection of the refrigeration unit.

* 4.2.4 Examination of preparation for delivery. An examination shall be made to determine that preparation for delivery as specified in section 5 is in accordance with MIL-R-12323.

4.3 Code and standard compliance. Proof of compliance with the requirements of UL, as applicable to 3.2.1, shall be available to the Government representative.

4.4 Methods of inspection.

* 4.4.1 Rigidity of structure. To determine compliance with the rigidity requirement of 3.4 for panel type mortuary refrigerators, a mortuary refrigerator shall be completely assembled and anchored at the base. A load of 500 pounds (227 kg) shall be uniformly distributed along the top side edge and this force should be applied in a horizontal direction to the top side of the refrigerator for 2 minutes and then removed. A 500-pound (227 kg) weight evenly distributed, shall then be applied to the top of the mortuary refrigerator in a vertical direction for 2 minutes and then removed. After tests, the refrigerator shall be examined for damage, permanent deflection, and distortion. Any permanent deflection, damage, or distortion shall be cause for failure of this test.

4.4.2 Strength of rack. The rack of a top compartment, with tray fully extended and locked, shall have a uniformly distributed load of 500 pounds (227 kg) applied slowly to the tray to determine compliance with 3.5.4, 3.5.5 and 3.5.5.4. The rack shall be unlocked, loaded tray operated to closed position, then reopened. Any nonconformance with 3.5.4, 3.5.5, and 3.5.5.4 shall be cause for failure of this test.

4.4.3 Operation test. To determine compliance with the applicable sections of 3.4, the refrigerator shall be precooled to not lower than 32°F (0°C) in a controlled ambient temperature of 110°F (43.3°C). A 4-hour operating test shall be conducted, recording refrigerated space temperature, operating time of refrigerating equipment, ambient temperature, load temperature, suction pressure, discharge pressure, and watt-hour input. The refrigerator temperature shall be maintained at 32°F to 34°F (0°C to 1°C). The full product load shall be simulated by a method recommended by the contractor and approved by the contracting officer. Any nonconformance with the requirements of 3.4 shall constitute failure of this test.

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4.4.4 Condensation test. The refrigerator shall be operated for a period of not less than 8 hours at a control setting that produces a cabinet interior temperature of 32°F, plus or minus 2°F (0 ± 1°C) in an ambient of 90°F, plus or minus 2°F (32 ± 1°C), and an ambient relative humidity maintained between 73 and 78 percent at all times. At the end of the 8-hour test the entire exterior of the refrigerator including door latch, hinges, and visible portions of the door gaskets shall be inspected for compliance with 3.5.6 for condensation visible to the eye or detectable by wiping with the fingers. Any nonconformance with 3.5.6 shall constitute failure of this test.

4.4.5 Refrigerant moisture test. To determine compliance with 3.6, during and after the operation test of 4.4.3, the moisture-liquid indicator shall be observed to indicate dry. If the moisture-liquid indicator shows that there is moisture in the system, drier replacement shall be made and the system operated until the moisture-liquid indicator shows dry.

4.4.6 Final leakage test. To determine compliance with 3.6, all refrigerator parts containing refrigerant shall be tested with a halide leak detector or other commercial method. Any leak shall constitute failure of this test.

4.4.7 Door seal test. To determine compliance with 3.5.6.3, the proper seating and sealing of door gaskets shall be tested at the four corners and midpoints on all doors by trying to insert a 0.005-inch (0.127 mm) leaf gage between the gasket and the seating face. It shall be observed that the insertion of the leaf gage shall be reasonably resisted. Any nonconformance with the requirements of 3.5.6.3 shall constitute failure of this test.

4.4.8 Density test for insulation. When rigid insulation is used, a 6-inch (152 mm) cube shall be cut and weighed to determine conformance to 3.3.4.3. When foamed in place insulation is used, one free foam sample of the insulation specified in 3.3.4.1, a minimum size 8-inch (203 mm) cube, shall be taken prior to each day's production. A 6-inch (152 mm) cube shall be cut from the center of the sample and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance to the density requirement of 3.3.4.1. Any nonconformance shall constitute failure of this test. In addition, a daily record of the foam insulation shall be kept as follows and made available to the contracting officer:

Calibration of mix (mix proportions)
Amount of flow (per box)
Time of flow (per box)
Mold temperature (per box)

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4.4.9 Electromagnetic compatibility tests. When electromagnetic compatibility is required, the refrigerator shall be tested by the contractor in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, and the Electromagnetic test report meet the requirements of MIL-STD-461.

4.4.10 Plastics tests.

4.4.10.1 Styrene plastic impact resistance test. Styrene plastic shall be tested for resistance to impact in accordance with method 1071 of FED-STD-406 to determine compliance with 3.3.6.1. Any nonconformance to the requirements of 3.3.6.1 shall constitute failure of this test.

4.4.10.2 Thermosetting plastics tests. Thermosetting plastics shall be tested for flexural strength in accordance with method 1031 of FED-STD-406 and for water absorption in accordance with method 7031 of FED-STD-406 to determine compliance with 3.3.6.2. Any nonconformance with 3.3.6.2 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Each refrigerator shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Each refrigerator preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323.

5.3 Marking. In addition to any special marking required by the contract, marking shall be in accordance with the applicable requirements of MIL-R-12323.

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6. NOTES

6.1 Intended use. The refrigerators covered by this specification are for mortuary use in Military or other Governmental Installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size and grade of refrigerator required (see 1.2).
(c) Electrical characteristics if other than specified (see 3.6 and 3.6.1).
(d) Type of compressor required (see 3.6 and 6.4).
(e) When electromagnetic compatibility is required (see 3.8).
(f) When condensing units are for overseas (see 6.4).
(g) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

6.3 Engineering design conference. Unless otherwise specified, at least 10 days prior to bid closing, the contracting officer should arrange an engineering design conference to discuss any questions concerning the details of Government drawings, instructions, and technical requirements of this specification. This conference is to assist the contractor in interpretation of the specification and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this specification nor relieve the contractor of his obligation to supply equipment specified.

6.4 Condensing units. Refrigerators for overseas shipment should have open type condensing units and refrigerators for use in continental United States should have thermally or mechanically sealed hermetic type condensing units (see 6.2).

6.5 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3)

6.7 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

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6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-2470E REFRIGERATORS, MECHANICAL, MORTUARY

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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